

ELECTRICITY CONSUMPTION ANALYSIS

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1. Introduction

Electricity is an important part of our daily life. Monitoring electricity consumption helps us understand how much electricity is being used and identify days with higher or lower consumption.

In this project, Python and Data Science techniques are used to analyze daily electricity consumption for one week. The data is stored in a simple dataset and analyzed using pandas. Matplotlib is used to create a graph for better visualization.

The main purpose of this project is to understand electricity usage patterns and demonstrate how Python can be used for basic data analysis.

2. Objectives

The main objectives of this project are:

1. To analyze daily electricity consumption.
2. To calculate total weekly electricity consumption.
3. To find the average daily consumption.
4. To identify the highest and lowest consumption days.
5. To visualize electricity consumption using a bar graph.
6. To understand how Python can be used for basic data analysis.

3. Tools and Technologies Used

The following tools and technologies were used in this project:

1. Python – Used for data analysis and calculations.
2. Pandas – Used to create and analyze the dataset.
3. Matplotlib – Used to create the electricity consumption graph.
4. Google Colab – Used to write and execute the Python code.
5. GitHub – Used to store and share the project and source code

4. Dataset Description

The dataset contains electricity consumption values for seven days of a week. It consists of two columns: Day and Units.

The Day column contains the names of the days from Monday to Sunday. The Units column contains the corresponding electricity consumption values.

The dataset used in this project is:

Monday – 8.2 units

Tuesday – 9.5 units

Wednesday – 7.8 units

Thursday – 10.1 units

Friday – 11.3 units

Saturday – 13.5 units

Sunday – 12.2 units

5. Methodology

The project was completed using the following steps:

1. A dataset containing daily electricity consumption was created.
2. The data was stored in a Pandas DataFrame.
3. Python was used to calculate total and average consumption.

4. The highest and lowest consumption values were identified.
5. The day with the highest consumption was found using Python.
6. A bar graph was created using Matplotlib to visualize daily electricity consumption.
7. The results were analyzed to understand the electricity usage pattern.

6. Results and Analysis

The analysis of the electricity consumption data produced the following results:

Total weekly consumption = 72.6 units

Average daily consumption = 10.37 units

Highest consumption = 13.5 units

Lowest consumption = 7.8 units

The highest electricity consumption was recorded on Saturday with 13.5 units, while the lowest consumption was recorded on Wednesday with 7.8 units.

The bar graph clearly shows the variation in electricity consumption throughout the week.

7. Data Visualization

A bar graph was created using Matplotlib to visualize the daily electricity consumption. The graph makes it easy to compare electricity usage on different days of the week.

The graph shows that Saturday had the highest electricity consumption, while Wednesday had the lowest consumption.

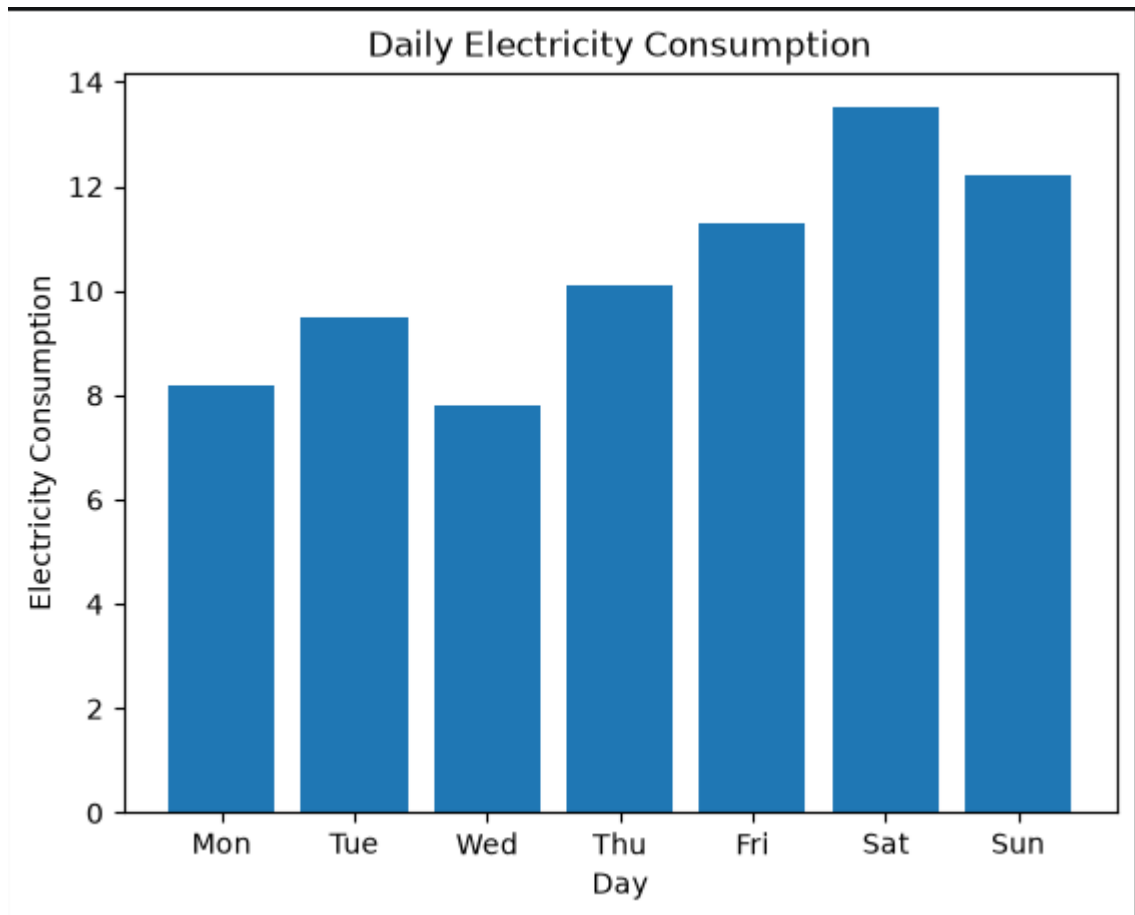


Figure 1: Daily Electricity Consumption

8. Conclusion

This project successfully analyzed electricity consumption for seven days using Python and basic Data Science techniques. The analysis calculated the total and average electricity consumption and identified the highest and lowest consumption days.

The results showed that Saturday had the highest consumption of 13.5 units, while Wednesday had the lowest consumption of 7.8 units. The bar graph helped to clearly visualize the variation in electricity usage.

This project demonstrates how Python can be used to analyze data and understand useful patterns from a dataset.

9. References

1. Python Documentation – <https://docs.python.org/>
2. Pandas Documentation – <https://pandas.pydata.org/docs/>
3. Matplotlib Documentation – <https://matplotlib.org/stable/>
4. GitHub Documentation – <https://docs.github.com/>